

1. Information about the indicator	
Goal	End hunger, achieve food security and improved nutrition and promote sustainable agriculture
Task	By 2030, end hunger and ensure access by all people, in particular the poor and people in vulnerable situations, including infants, to safe, nutritious and sufficient food all year round
The indicator	Prevalence of undernourishment
2. Information about the organization	
Organization	Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan
3. Definitions and concepts	
Definition	The prevalence of malnutrition (PoU) is an estimate of the proportion of the population whose usual food intake is insufficient to meet the need for dietary energy necessary to maintain a normal, active and healthy life. This estimate is expressed as a percentage.
Basic concepts:	Malnutrition is defined as a condition in which a person has regular access to an amount of food that is insufficient to provide them with the energy they need to lead a normal, healthy, and active life, taking into account their own dietary energy needs. Although, strictly speaking, the concept of "malnutrition", as defined here, is different from the physical condition of "malnutrition" and "malnutrition", as it refers to a condition caused by insufficient food intake, and not to the result in terms of nutrition. In French, Spanish and Italian, the difference is reflected in the use of the terms food, alimentación, or alimentazione, instead of food, nutrición or nutrizione, in the name of the indicator. A more appropriate expression in English, which would make the meaning of this indicator more accurate, could be "the prevalence of malnutrition," but the indicator has long been associated with the term "malnutrition." While the condition of malnutrition applies to individuals due to conceptual and data-related considerations, this indicator can only apply to a population or a group of individuals. Thus, the prevalence of malnutrition estimates the percentage of individuals in the group who are in this condition, but it does not allow us to identify those individuals in the group who, in fact, suffer from malnutrition.
Justification and interpretation	The indicator has been used by FAO to monitor the World Food Summit goals and MDG 1C targets at the national, regional and global levels since 1999. This indicator allows you to track trends in the degree of insufficient calorie intake of the population over time, resulting from a combination of changes in the general availability of food, in the ability to access it in households, in the socio-demographic characteristics of the population, as well as differences between countries and regions at any given time. The parametric approach adopted by FAO makes it possible to obtain reliable estimates for relatively large populations. Since it reflects a severe condition in the absence of food, it is fully consistent with the spirit of the Goal, which aims to reduce hunger.
4. Data sources and collection methods	

Data sources Data	Sample survey of households by standard of living.
collection methods	Sample survey of households by standard of living.
Unit of measurement	Percent
Periodicity	Annual
5. Calculation method and other methodological considerations	
Calculation method:	Calculation method: The indicator is calculated at the population level. To this end, the population is represented by an "average" individual, for whom the probability distribution of the usual daily calorie intake is modeled by a parametric probability density function (pdf). Once the parametric probability density function is characterized, the indicator is calculated as the integral probability that the daily habitual calorie intake (x) is not lower than the lower limit of the range of normal energy nutritional requirements for this representative or the average individual (MDER), as in the formula below: $PoU = \int_{MDER}^{\infty} f(x DEC; CV; Skew) dx$ where DEC, CV and Skew have an average value, the coefficient of variation and asymmetry characterize the distribution of habitual levels of calorie intake among the population. Until 2012, the probability distribution f(x) was modeled as a logarithmically normal parametric probability density function based on only two parameters: the arithmetic mean and the coefficient of variation. In its latest development, it is modeled as a parametric probability density function of three parameters, capable of representing various degrees of skewness, ranging from a symmetric normal distribution to the positive skewness of a logarithmically normal distribution. Flexibility in capturing different degrees of asymmetry is necessary, given the fact that human energy consumption is naturally limited by physiological norms. Therefore, it can be assumed that average consumption levels are increasing, and the distribution asymmetry is decreasing, gradually moving from a logarithmically normal distribution (with positive asymmetry) typical of a population where average food consumption is relatively low to (symmetrical) normal distributions. The normal skewness and logarithmic normal skewness of the distribution family are allowed to describe all possible intermediate degrees of positive skewness. http://www.fao.org/3/a-i4046e.pdf detailed description) A custom R function is available from the FAO Statistics Department to calculate estimates of the prevalence of malnutrition, taking into account four parameters for estimating the level of calorie intake (DEC), coefficient of variation CV, Skew and minimum calorie intake (MDER). Different data sources can be used to estimate different model parameters. Calorie intake level (DEC) The average value of the distribution of calorie intake levels for the average individual from the population (DEC) on average corresponds, by definition, to the daily level of food consumption of the population per

person.

DEC can be estimated based on food consumption data obtained from surveys that are representative of the population group of interest. Depending on the survey structure, they can be used to assess DEC at the national and subnational levels, or by geographical area and socio-economic groups of the population. Unfortunately, although the situation is rapidly improving, representative surveys collecting data on food consumption are still not available for every country and cannot always be conducted every year.

For the country's population, only DEC can be calculated based on information about the total supply and use of all food products in the country, where the contribution of each product to the availability of food for human consumption is expressed in the energy value of nutrition, and their sum divided by the population. The main source of data on national food balances is the Food balances (FBS), which are maintained by FAO for most countries in the world (see <http://www.fao.org/economic/ess/fbs/en/>), based on official data provided by the member countries, and distributed through FAOSTAT (http://faostat3.fao.org/download/FB/*/F)

Coefficient of variation (CV)

Surveys that contain information on food consumption at the individual and family levels are the only source that is able to directly assess the coefficient of variation in CV of usual food intake for an individual representative of the population. Unfortunately, the survey data on food consumption is fraught with many problems that complicate the reliable estimation of the CV coefficient of variation.

In principle, repeated observations of daily intake for each individual in the sample will be necessary to assess the level of habitual consumption and to determine measurement errors. In addition, information should be collected from the same individuals or households at different times of the year to account for possible seasonal fluctuations in caloric intake levels. Due to their cost, national surveys on individual food consumption with such characteristics are very rare, and virtually non-existent in most developing countries. As a result, the most common data sources for estimating the CV coefficient of variation are multi-purpose household surveys, such as studies of criteria for assessing living standards, household income and expenditure studies (or household budget surveys), which also collect information on food consumption. When using data collected at the household level, it is necessary to pay close attention to the different levels of purchases or receipt of food from the level of their actual use (consumption and irrational use) during the specified reporting period and taking into account the correct recording of the number of persons who participate in consumption; in addition, at the household level, the data will hide the variability of-for the intra-family distribution of food.

For all these reasons, the coefficient of variation calculated on the basis of the average daily calorie intake per capita recorded for each household included in the survey is not a reliable estimate of the coefficient of CV variation, which should reflect differences in the levels of habitual (rather

than random) daily calorie intake on an individual (rather than household) level. Empirical estimates of the CV coefficient of variation from household survey data are overstated due to false variability caused by measurement error, differences between random and habitual consumption, differences between acquisition and actual consumption, and seasonality; in addition, they do not reflect the variability in caloric intake of the population related to the individual characteristics of family members (for example, gender, age, body weight, and physical activity level).

Thus, when using data collected through household surveys, the CV variability coefficient is best estimated indirectly, taking into account false variability, and adjusted for inter-individual (along with inter-household) variability. The easiest way to start is to classify households into homogeneous groups and calculate the coefficient of variation in the average daily calorie intake per capita of different household groups. This leads to an estimate of the inter-bit component of the CV coefficient of variation, labeled CV_H. The assessment of the interindividual component of the CV coefficient of variation, labeled CV_I, is applied to each population group based on its composition by gender, age, and body weight, and these two components are combined to obtain the necessary assessment.:

$$CV^{\wedge} = \sqrt{[(CV_H)^2 + (CV_I)^2]}$$

When there is no household survey data for countries and years, an indirect estimate of the CV coefficient of variation, CV_IND, is obtained using regression, which reflects the values of GDP per capita, the Gini coefficient for income, and the relative food price index (FPI) for the CV coefficient of variation, while taking into account regional shifts (REG).

$$CV^{\wedge}_IND = \beta_0 + \beta_1 \text{ GDP} + \beta_2 \text{ GINI} + \beta_3 \text{ FPI} + \beta_4 \text{ REG}.$$

Regression coefficients are estimated based on a set of data and years for which information on the coefficient of variation CV, GDP (GDP), Gini coefficient (GINI) and the relative food price index (FPI) is available.

Asymmetry:

Since the asymmetry is not strongly affected by the presence of false variability, the asymmetry is estimated directly from the data of average daily calorie intake at the household level with a single exception, eliminating rare extremely high or extremely low values. If the empirical skewness coefficient exceeds the value that would correspond to the skewness of a logarithmically normal distribution with a given mean and coefficient of variation, the parameter is ignored and the logarithmically normal distribution of the two parameters is used for f(x). (See. <http://www.fao.org/3/a-i4046e.pdf> for more information).

Minimum caloric intake (MDER)

Human energy needs are calculated by multiplying the regulatory requirements for basic metabolism (BMR, per kg of body weight) by the ideal weight of a healthy person of a given height, and then multiplied by the physical activity coefficient (PAL). Thus, the ranges of normal energy requirements are calculated for each gender and age group of the

population, noting that there are a number of body mass index (BMI) values – from 18.5 to 25 – that correspond to the state of health. This means that any resulting increase can correspond to a range of healthy body weight values, and therefore a range of values for the amount of energy required for basic metabolism (BMR).

Taking into account the information about the average height and in view of the fact that there may be people in the group who have different levels of physical activity, the minimum, average and maximum energy nutritional requirements can be calculated for each gender and age category, taking into account a special height allowance for people aged 0 to 21 and for women during pregnancy and lactation.

(See <http://ftp.fao.org/docrep/fao/007/y5686e/y5686e00.pdf> for more information).

The minimum caloric intake (MDER) of a certain population group, including for the population of the country, is obtained as a weighted average of the minimum ranges of energy needs of each gender and age category, using the population in each class as weights.

When calculating the prevalence of calorie deficiency among the population, there was often confusion between the concepts of minimum calorie intake (MDER) and recommended calorie intake, and about the appropriate threshold for calculating the probability of deficiency. The reason why the probability of calorie deficiency should be calculated with reference to the minimum calorie intake (MDER), rather than ADER (which, on the contrary, can be used as an estimate of the average recommended food intake for the entire population) is that you just need to recognize the fact that in any population there is a certain range of normal variability in needs; Using ADER as a threshold indicator will greatly overestimate malnutrition, since the proportion of the healthy population that consumes less than the average level will also be taken into account, simply because their needs are below the average level. If necessary, the ADER, or the average recommended caloric intake among the population, should be used to calculate the difference in caloric intake.

Disaggregation:

Due to the reliance on national food balance data to estimate the average calorie intake of the population, the global monitoring of MDG 1C and the goal of the World Food Summit is based only on estimates of the prevalence of malnutrition at the national level.

In principle, this indicator can be calculated for any particular population group, provided that sufficiently accurate information is available to characterize the parameter models for this group, that is, if there is information about the group's food intake, age/gender structure, and possibly the level of physical activity.

Thus, the framework for disaggregation depends crucially on the availability of surveys designed to be representative of subnational populations. Given the established practice in developing national household surveys, sufficiently reliable information is rarely available for disaggregation in addition to the level of the macro-region of residence (urban-rural) and the main regions/districts in the country. Since most of

the surveys used are designed to accurately measure income distribution, a conclusion can be drawn based on an assessment of the prevalence of malnutrition among different segments of the population. Gender disaggregation is limited by the ability to identify and group households by gender (for example, the gender of the head of household, or the ratio of men to women).

- Processing of missing values:
- At the country level

In the absence of data on food consumption from recent household surveys, the modeled estimate of the prevalence of malnutrition (PoU) is based on an estimate of calorie intake (DEC) from Food Balance Sheets, an indirect estimate of the coefficient of variation (CV) based on information about a country's GDP, the Gini coefficient for income, and the relative price index. on food or other indicators of the country's development such as, how is the mortality rate of children under 5 years of age and the assessment of the minimum caloric intake (MDER) based on data from the World Demographic Perspectives of the United Nations Population Division.

- See the section on calculation methods for more details.

At the regional and global levels

Missing values for individual countries are implicitly imputed as equal to the weighted average of the calculated population values for countries that are in the same region.

Regional indicators:

Regional and global indicators of the prevalence of malnutrition (PoU) are calculated as: $PoU_REG = (\sum_i PoU_i \times N_i) / (\sum_i N_i)$

where PoU_i is the prevalence of malnutrition (PoU) for all countries in the region for which data are available to calculate a reliable estimate, and N_i is the corresponding population.

Sources of discrepancies:

Many countries have produced estimates and reported on estimates of the prevalence of malnutrition (have produced and reported on estimates), including in their national MDG reports, but almost always used a different methodology than that developed by FAO, which makes national figures incomparable with those reported by FAO for global monitoring.

The most common approach used in the preparation of national reports was to calculate the proportion of households for which the average daily calorie intake per capita would be below the thresholds based on the daily recommended dietary intake, usually set at 2,100.00 kcal, based on household survey data. In some cases, lower thresholds of about 1,400.00 kcal were used, probably as a reaction to the fact that the proportion of households reporting an average daily intake of less than 2,100.00 kcal per capita were implausibly high estimates of the prevalence of malnutrition.

Almost without exception, excessive variability in calorie intake data is

	not considered, and reports show limited or no progress in reducing the prevalence of malnutrition over time. As already mentioned in the section on the calculation method, the results obtained through these alternative methods are highly unreliable and are almost certainly biased towards overestimation. It is therefore advisable to make a concerted effort to promote the use of FAO methods also in the preparation of national reports. FAO stands ready to provide all necessary technical support.
Comments and restrictions:	
6. Data availability and disaggregation	
Data availability and gaps:	The data is available on the website of the Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan.
The level of disaggregation:	-
7. Comparability with international data/standards	-
8. References and documentation	www.stat.gov.kz